

Semantic Model for Adaptive Human-Agent Dialogues

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Abstract — A common conversation between an older adult and a nurse about health-related issues includes topics such as troubles with sleep, reasons for walking around nighttime, pain conditions, etc. Such dialogue can be regarded as a "natural" dialogue emerging from the participating agents' lines of thinking, their roles, needs and motives, while switching between topics as the dialogue unfolds. The purpose of this work is to define a generic model of purposeful human-agent dialogues suitable for health-related topics. This is done based on analyses of scenarios, personas and models of human behavior. The results include four models, which need to be included in a software agent's belief base; i) a user model, ii) a model of the domain knowledge related to the topic of the dialogue, iii) an agent model, and iv) a dialogue activity model. The models were implemented into a prototype system for human-agent dialogues, which was evaluated by therapists and a group of older adults.

Keywords Personalization · Dialogue systems · Health promotion · Argumentation · Intelligent agent · User model

1 Introduction

Assistive technology aims to support an individual in accomplishing activities, which they need to be able to do in the presence of decreased functionality or ability. In this work the definition of *active* assistive technology used by Kennedy and co-workers [1] is applied to distinguish systems, which includes automated processing of health information during a human agent's interaction with the system and which may output tailored responses to the human

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agent in the process. The purposes of active assistive technology include the following: to increase knowledge, assist in deciding about actions to make, and promote changes of unhealthy behavior (i.e., in the form of *behavior change systems*).

Our work focuses on dialogues between a human actor and an active assistive technology in the form of an intelligent software agent. The concept of Embodied Cognitive Agents (ECA) is generally used for such systems. The ECA uses a virtual representation of a human with the ability to send information through body language in addition to linguistic messages. However, we restrict the focus in this work to structured linguistic dialogues based on semantic models of relevant knowledge.

In this article, a dialogue system is presented, which enables a human actor to conduct a dialogue with a system agent for the purpose of handling health-related issues in a home environment. The aim is to enable the dialogues between a human agent and a software agent, which are tailored to the human's needs, preferences and goals. For this purpose, the software agent needs the following capabilities, which has to different extent also been described in literature (e.g., [2–5]):

1. Being autonomous (e.g., decides upon actions to make, takes initiatives in order to reach a goal),
2. Utilize a *semantic model* shared between the human and the agent for communication and knowledge exchange purposes,
3. Handle knowledge obtained from different heterogeneous sources, e.g., in a home environment,
4. Reason and make decisions in the presence of uncertain and incomplete information,
5. Generate new knowledge and learn,
6. Being cooperative,
7. Being able to deal with affective components and topics in a dialogue,
8. Being adaptive to the human's needs, preferences and the situation.

The goal is to develop and combine the above-mentioned capabilities to build a software agent that interacts with the human as their personal coach, friend or a discussion partner, i.e., as a *Coach Agent* as described in [6].

Typically, a team of healthcare professionals can be involved in an elderly person's care, such as doctors, nurses and occupational therapists. These together with the human cooperate to gain optimal understanding of the human's health condition. The support provided to the human is aimed to be person-centered. Consequently, there is a need for combining information from different sources in order to understand the external factors affecting the human. We deal with a complex environment where there are multiple sources of information to consider for an agent: 1) the information about the human agent's daily activities observed by an activity recognition system at home, 2) the information about the human agent's medical health condition obtained from the human agent, domain professionals and relatives, and 3) the human's

preferences, obtained initially as a part of the baseline assessment, which, however may change later.

In this paper, we focus on the knowledge, which the software agent needs to have and obtain for contributing to the dialogue, and for behaving in a purposeful way. The main contribution of this paper is a conceptual and formal model of the Coach Agent's knowledge including a model of the dialogue activity, which enables a purposeful interaction between the human agent and the Coach Agent in the pursue of understanding and improving the human's health condition.

An initial prototype dialogue tool is implemented and integrated in a support application, which enables the human actor conduct a dialogue with the Coach Agent in their home environment. The prototype was evaluated in a pilot study involving a group of older adults and a group of therapists.

This paper is organized as follows. In the following section the methods are described and in Section 3 an analysis of related work is provided. Section 4 introduces the use case scenario of an older woman and the results of the analysis of the scenario, which provided requirements and motivations for our design. In Section 5 a model of the dialogue activity is provided and in Section 6 a model of the Coach Agent is provided. Section 7 describes the pilot evaluation study. The results are discussed and the article ends with conclusions and comments on future work.

2 Methods

A literature study was conducted where the following topics were studied: i) the agent's role, ii) the purpose of dialogue systems, iii) the knowledge model of the agent, the user, the domain and the dialogue activity, iv) who develops the knowledge model, v) sources for the knowledge, vi) representation format and how generic the representation is, and vii) issues regarding the dialogue execution.

The semantic model for adaptive human-agent dialogues is designed based on the persona and scenario of a female older adult named Rut, and the dialogues aimed for supporting a human actor described in [6, 7]. The scenario was analyzed, providing baseline requirements and a conceptual model of goal-directed activity.

The theoretical base for analysis is Activity Theory [8], which also informed the models generated as results of our work.

A pilot evaluation study was conducted involving a group of five female professionals in occupational therapy and physiotherapy, specialized in the needs of older adults, and a group of eleven older adults, six women and five men. The study was formative, with the results aimed to inform further development.

The main research questions, which were targeted by the evaluation study related to the overall idea of a dialogue system for supporting everyday issues,

how a sense-making dialogue would unfold and what topics would be interesting to elaborate upon. Other research questions related to interaction design of the dialogue system.

The evaluation study was limited to the initiation of inquiry dialogues and the conduction of nested information-seeking dialogues, in the context of the use case scenario of Rut. The information about Rut was forming the base for the Coach Agent’s user model and the priorities of goals and tasks. This information and consequently, the user model changed during the testing, depending on how the participants interacted with the system.

Observation of use and interviews were the methods used for collecting data. The older adults and two of the therapists were individually given the task to play the role of the persona in our scenario, other person or themselves, select topics of interest to initiate a dialogue and responding the way they wished.

The user was also asked to evaluate the appropriateness of each statement of the Coach Agent within its context where it was stated by marking one of four alternatives on a Likert scale, presented in connection to each of the Coach Agent’s statements. A purpose of the pilot study was also to evaluate this method.

The system and the scenario was demonstrated to the remaining three therapists in a focus group session and discussed.

The data was analyzed qualitatively using content analysis.

3 Related Work

Research literature shows an increasing number of applications of software agents that interact with the human actors for health-related purposes (e.g., [9, 10, 5, 11–13]). A review of behavior change systems utilizing personalization technologies for accomplishing active assistance is provided in [1]. It was concluded that an agent provides an effective interface modality especially for the applications that require repeated interactions over longer period of time, which is crucial for applications supporting behavior change [1, 14].

One example is the system, which has been developed for older adults with cognitive impairment described in [9]. However, it focuses mainly on generating reminders about the activities of their daily living and takes no part in a complex dialogue with the user. Agent-based systems which interact with human actors through dialogues are less common. They are typically developed for a specific task or for a limited domain, and the dialogues are tested in specialized environments [13, 11].

Bickmore et. al [13] developed an animated conversational agent that provides information about a patient’s hospital discharge plan in an empathic fashion. The patients rated the agent very high on measures of satisfaction and ease of use, and also as being the most preferred over their doctors or nurses in the hospital when receiving their discharge information. This was

related to the agent capability to adapt to the user's pace of learning and giving information in a nonjudgmental manner. Another reason for the positive results was the amount of time that the agent spent with users helped them to establish a stronger emotional bond.

In the counseling dialogue system discussed in [14], the agent performs motivational interviewing with its knowledge about behavioral medicine and expresses empathy in dialogues. It provides the human agent with advice about exercise and healthy actions and follows up by asking the human followup questions. However it limits its services to counseling only and does not reason with evidence about the human's change of behavior based on information about their activities.

The agent-based approach to diagnostic decision-making presented in [15] has the potential to support the medical professionals in improving their diagnostic assessments of patients. The purpose is to guide the user in the diagnostic reasoning and support learning and a change of behavior. However, the user's active participation in the dialogue is limited.

To summarize, the most common Agent roles, and consequently, the purposes of the exemplified dialogue systems are the following: i) promoting the human actor to maintain control, learn and change behavior, ii) to guide in reasoning, decision-making and other activities, and iii) to inform in an emphatic and patient way, and behave in a way tailored to the human agent's pace, needs and preferences.

The canonical model of an agent described by Fox and coworkers [3] is an extension of the Belief-Desire-Intention (BDI) model of an autonomous agent (e.g., [2]) and provides formal semantics for the concepts argumentation, decision-making, action, belief, goal (desire) planning (intention) and learning for agents. Literature describes implementation in systems, however, mainly for multi-agent reasoning and decision making without active participation of a human agent in the process [3]. To our knowledge, this model has not served as a base for dialogues between software agents and human agents beyond simple alerts or providing decisions.

The architecture of the agent-based ambient assistive living system presented in [6] integrates knowledge repositories, developed using ACKTUS (Activity-Centered modeling of Knowledge and interaction Tailored to USers), which is a platform for end-user development of knowledge-based systems in the medical and health domain. These repositories are built on semantic models represented using RDF/OWL, which were in [16] extended to integrate components vital for enabling agent dialogues, based on the canonical model for agent architectures presented by Fox and coworkers [17]. This work lead to a multi-agent system, which simulates a dialogue between an agent representing a physician who is the user of the system and who selects a diagnostic hypothesis in a patient case, and an agent representing a domain expert [15]. The domain knowledge and patient knowledge was used for creating arguments, based on sources with different strengths, which the agent-system evaluated for providing the user support in diagnostic decision-making. The

domain knowledge was in this case retrieved from clinical practice guidelines, consensus guidelines, and other medical sources.

The ACKTUS core ontology was further extended to incorporate information about a user's environment, for the purpose of forming a user model, which an agent can use for adaptation of support in daily activities [18].

An initial design and implementation of an agent-based dialogue system where an agent communicates with the end user in a home environment on health-related topics is presented in [19]. The agent's belief base was created by obtaining information from the ACKTUS knowledge repositories, which had been created by domain experts in the rehabilitation domain [7]. The knowledge consisted of both factual knowledge related to a user, which was used for creating a baseline user model, and procedural knowledge, in the form of rules obtained from domain knowledge repositories. Moreover, interactive templates, or protocols for assessment of different aspects were retrieved from the domain knowledge repositories. The knowledge had been structured and implemented by the domain experts to be used as assessment protocols by therapists in clinical interviews, or in human-agent dialogues.

The counseling dialogue system discussed in [14] also uses an ontology to represent the user model and the domain knowledge. The domain knowledge is based on behavioral medicine. Their approach consists of six models: theory model, behavior model, protocol model, user model, external data model and a task model. The authors have themselves developed the ontology based on a review of concepts in behavioral medicine, prior work, and discussions with experts in behavioral medicine and computerized interventions. Their task model consists of the dialogue activity information with dialogue actions. However, the domain experts are not directly involved in the modeling of the agent's knowledge in this system therefore it lacks features for modifications such as future changes in domain knowledge and also it lacks the representation of goals which drives the behavior of the dialogue system.

Only a few applications developed for behavior change integrated into the system semantic models grounded in theories relevant for behavior change [1].

In argumentation literature there are several theoretical frameworks, which aim to formalize dialogues of different types for multi-agent systems. Typically, these adopt the approach of *dialogue games*, since they aim at selecting a "winner" of a dialogue, and then the game rules aims to create the dialogue game a fair competition (e.g., [20]). Based on argumentation schemes, dialogues can be built, e.g., for reasoning about actions to make related to the value of conducting a particular action [21]. Theoretical frameworks have been formalized also for handling nested dialogues of different types (e.g., [20]). However, these are typically not tested with dialogue information from real situations, and consequently, their applicability in situations such as our use case remains to be evaluated.

The base for interaction in [19] was implemented as information-seeking dialogues through which the user can inform the system about preferences, priorities and interests. Based on this information, the system provides feedback to the user in the form of: 1) suggestions of decisions, 2) advices and

3) suggestions of actions to make for obtaining more knowledge about a situation. However, in order to accomplish other types of dialogues, which are less structured than the *information-seeking* dialogues, an agent architecture is needed, which allows the agent to plan the moves based on e.g., the purpose of the dialogue, the situation, the knowledge available, etc. Therefore, this paper extends earlier work, by developing the components for constructing an intelligent software agent that interacts with a human in a more autonomous, adaptive and purposeful way.

4 Model Based on a Scenario

Our persona called Rut [6] shares similarity with some of the participants in a study conducted by Lindgren and Nilsson [22], and is therefore considered representative. Rut is 84 years old, suffers from pain in her back and legs and had suffered from few falls before a hip fracture. We envision that Rut begins to walk around nighttime, and that she may discuss the situation and her sleep with a nurse. The nurse asks a few specific questions about Rut's activities and health, and the dialogue will wander from one aspect to another, sometimes coming back to a topic already mentioned.

This example of a natural dialogue is rather different from dialogues described in literature, which aims at reaching one particular goal of a dialogue, e.g., [23]. Following the categorization of dialogue types, described by Walton and Krabbe [24], the dialogue with the nurse is a simplified example of a combination of different goals: finding information (*information-seeking* type), generate new knowledge, i.e., conclusions (*inquiry dialogue* type) and deciding upon actions to make (*deliberation* type). In case one of the agents has reasons for arguing for one particular action to be made with the purpose to convince the other, e.g. for safety reasons, the dialogue may include a *persuasive* part, e.g., to convince Rut that she needs to go to the hospital for investigation. Based on this, we can define a set of *generic goals* for the agent to use in its organization of dialogues. More concretely, the generic outcome of each type of dialogue is the following: *information*, *new derived knowledge*, *plan of actions* and a *change of priority*. It may be that all of these types of dialogues need to be conducted in order to fully explore a particular *topic*. These generic goals will correspond to actions defined in Section 5 and *schemes* described in Section 5.2.

Consequently, the agent needs to be able to handle nested multi-purposed dialogues with different topics. In order to accomplish this, the agent needs to be able to distinguish between *topic*, *generic goal* and have a semantic model of how these inter-relate in a particular situation. For instance, if the agent would have the dialogue with Rut instead of the nurse, the agent needs a semantic model for how walking around nighttime relates to sleep patterns, pain, cognitive ability, medication, worries, etc. (i.e., a *domain model*). Moreover, it needs strategies to plan next moves, based on a knowledge model, which may not provide a pre-defined hierarchically organized plan of actions

based on goals and sub-goals to be followed, but rather a collection of prioritized actions, among which the order may become determined and changed by the dialogue involvement and Rut's line of thinking (enabling *situatedness* of dialogues and the agent to adapt).

The *topics* relate to the human agent's goals and priorities. From the software agent's perspective, goals relate to finding answers to questions, finding and deciding about actions to make in order to increase levels of satisfaction, evaluate actions made, etc. Consequently, for the human agent, goals are purposeful, topic-driven and context dependent.

Therefore, the agent in our approach combines a generic goal with a specific topic, in order to identify the *specific goal* with an action. In the following we give an example of a dialogue, where the topic selected by the human agent is "Sleep pattern may be disturbed", which is a claim representing a belief, which has not been verified. This is an example of the initiation of an *inquiry* dialogue, aimed at finding and verifying new knowledge. The agent can choose to conduct a dialogue with the purpose to find information about sleep (specific goal) and initiates an information-seeking dialogue. Topics related to sleep are e.g., pain and medication, and consequently, in this situation, the sub-goals are to *find information about pain* and *medication*, respectively. This is an example of how different types of dialogues can be nested, for the purpose to feed information and knowledge into the cooperative reasoning process of solving the overall topic, here chosen by the human actor.

The following is an example of a part of a dialogue, generated by the prototype system in collaboration with Rut, where CA represents the Coach Agent:

1. Rut: Hi Coach, I wonder if not my "Sleep pattern may be disturbed".
2. CA: Hi, Rut. Okay, then I would like to ask about *sleep*. Did the sleeping pills work for you tonight?
3. Rut: No.
4. CA: Why did you not sleep well last night?
5. Rut: Needed to pee and pain.
6. CA: Now I would like to ask about *pain*. Where do you have pain?
7. Rut: In my back.
8. CA: How severe is your pain?
9. Rut: Severe.
10. CA: Let us go back to *sleep*. How often do you need to get up each night?

To summarize, the scenario emphasizes the need for the following three semantic models for enabling human-agent dialogues: 1) a *domain model*, which contains generic knowledge about a particular domain, 2) a *user model*, which contains the collected knowledge about the human agent, and 3) a *dialogue activity model*, which provides the relations between topics, generic and specific goals and actions to be made by the participating agents in a dialogue. Moreover, the third model can be seen as the generic behavior knowledge model of the Coach Agent, however, enriched with specific knowledge, which relates

only to the Coach Agent, forming the fourth model: 4) an *agent model* (Figure 1).

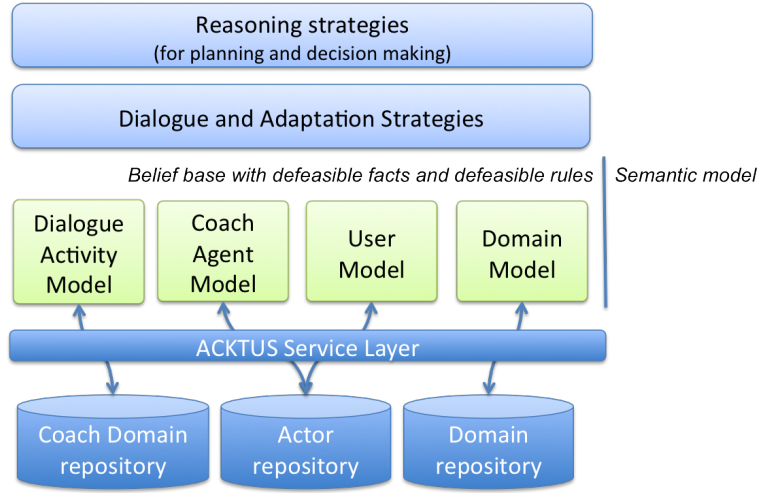


Fig. 1 The agent architecture, containing the Coach Agent's belief base consisting of four models, and its different sources of knowledge.

Finally, the need for *strategies*, which enable dynamic dialogues following the human agent's line of thinking is illuminated by the scenario.

The user and domain models are extracted from the ACKTUS ontologies [25] in the dialogues between human and software agents. The dialogue activity model and a model of the Coach Agent is developed as a part of this work, and presented in the following sections.

5 A Model of the Dialogue Activity

As described in Section 4, and following the second requirement listed in the Introduction, the software agent needs to share a common semantic model with the human agent, in order to be able to reason and decide upon which actions are valuable to the human agent. In this section the dialogue activity model will be defined.

The activity theoretical model of human activity was used for organizing actions and their goals at different levels [8]. Activity theory captures the complexity in human activity, including human *needs and motives* as driving force, *goal-directed actions*, and *operations*, which constitute basic actions conditioned by the agents and the environment.

Figure 2 provides an overview of the shared model of a dialogue activity in which both a human and software agent participate. As can be seen, in the dialogues a common *topic* is the representation of the overall *motive for*

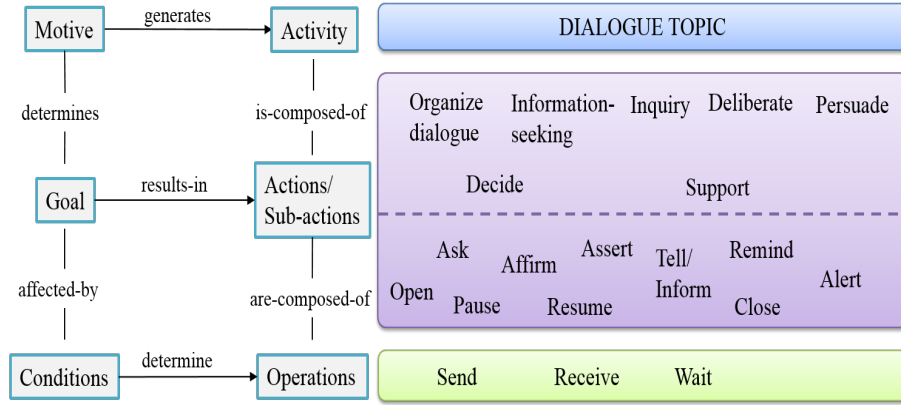


Fig. 2 A model of the dialogue activity, which both a human and a software agent are expected to comply with.

activity. Each dialogue is initiated by one of the agents, by posing a selected topic to the other agent.

The identified operations relate to passing information to and receive information from the other agent(s) (*send* and *receive* in Figure 2), and wait for responses.

In addition to these basic and top levels of activity, a set of potential generic sub-actions has been defined, which may take place in the conduction of a dialogue. These are also organized in a hierarchical representation, since an action may serve as a sub-action to more than one action at different levels. However, we identify the following generic set of actions at the highest level: *information-seeking*, *inquiry*, *deliberative*, and *persuade*, related to the common goals of different types of dialogues described in Section 4: seek information, find new knowledge, decide upon actions to be made and lead the other agent to change opinion. To these actions, we add the action *organize dialogue*, which contains the sub-actions typical for multi agent dialogues, e.g., *open* and *close*. These sub-actions are called *moves* in multiagent literature [23]. We extend the set of moves (i.e., sub-actions) and include the following moves as valid actions for the agent to take as part of the different actions: *open*, *close*, *pause*, *resume*, *ask*, *assert*, *affirm*, *inform*, *remind* and *alert*.

The model of dialogue activity presented in this section defines the actions, which need to be common between the participating agents as a part of a common semantic model. In addition to these, there are tasks, which are conducted by the agent "internally" during a dialogue, which also relates to the execution of a dialogue. These tasks are aimed to handle interruptions of dialogue, cases when the human agent does not respond, etc. Other tasks are creating and updating the agent's own belief base, decide about what move to make next, which dialogue to initiate, etc. A description of these tasks included in the generic organization of a dialogue mentioned in Figure 2 is provided in [26], while the tasks included in the collaborative decision-making conducted as a

part of human-agent dialogues are described in the following sections (Figure 1). In order to distinguish the lower level actions, which represent different types of moves in a dialogue, we hereafter call these *moves*, and denote the different dialogue types at a higher level as *actions*.

5.1 Dialogue Moves

Formally, a dialogue move for the agent in this work, is a tuple (t, a, m) where t is timepoint of the dialogue move, a is the agent and m is the dialogue move. The set of dialogue types includes the following: information-seeking (*is*), inquiry (*wi* or *ai*), deliberate (*dd*), persuasive (*pd*) and support dialogue (*sd*). Based on this tuple, we define the actions as follows (Table 1).

Table 1 Valid actions, i.e., moves and their formats. All moves contain the time identifier t_n and a_i , the identification of the agent, which performs the move.

Move	Form	Comment
Open	$(t_n, a_i, open(d_o, \alpha_m))$	α_m is the topic of a dialogue
Close	$(t_n, a_i, close(d_o, \alpha_m))$	α_m is the topic of a dialogue
Ask	$(t_n, a_i, ask(CQ))$	CQ is a structured question
Affirm	$(t_n, a_i, affirm(\alpha_m))$	α_m is a confirmative expression
Tell	$(t_n, a_i, tell(\alpha_m))$	α_m is the message, typically an advice or information
Remind	$(t_n, a_i, remind((\alpha_m, G), a_j))$	α_m is the reminder, G is the set of reasons for the reminder, and a_j is the agent targeted for the reminder
Alert	$(t_n, a_i, alert((\alpha_m, G), a_j, t_l))$	t_l is the timeout for the action
Assert	$(t_n, a_i, assert(\alpha_m, G))$	α is the claim and G is the set of grounds for the claim

The *Open* move is the first action carried out to initiate a dialogue, and the *Close* move is used for closing, or stating the end of a dialogue. *Tell* is the simplest form of action that an agent makes. The *Remind* action is used by the agent to remind the human to do an action. The *Alert* move is also similar to the remind action but with a timeout. It recommends the human to take an immediate or urgent action for a critical situation, for example, if Rut had forgotten to take medication after breakfast then send alert to take medicine. When an activity needs to be urgently completed, then a timeout is attached with that activity so that the reminder can be repeated in different ways. In this work, when the agent needs to obtain information from the human actor, it uses the *Ask* move. It fetches the relevant questions to be asked from the ACKTUS knowledge repository and stores them in its knowledge base. The *Assert* move is used for making a claim about some topic, and it is supported by the set of grounds G on which the claim is being based. The claim is a defeasible fact.

The *Affirm* move is used to acknowledge the other agent and its expressions in a more generic way. The typical purpose is to make the other agent comfortable, providing a "fill" in the dialogue.

5.1.1 Rules for Dialogue Moves

There are rules related to how the moves can be applied. In dialogue games, the restriction that agents take turn in the dialogues, and are allowed to perform only one move at the time does not apply in our dialogues. The reason is that the agents are not competing for "winning" the dialogue, and natural dialogues do not follow this restriction.

Other restrictions apply, such as the open move for a particular dialogue needs to take place before a close move. Similarly, in case a pause move is done, no move can be done within this dialogue until the dialogue is resumed by a resume move.

Only one dialogue can be active at a particular time point. This means that when a new dialogue is opened by an open move, and there is an ongoing dialogue, this is paused.

In order to close a dialogue, both agents need to agree to close, which is done by a close move by each agent.

Some moves have the purpose to defer the responsibility for acting to the other agent. Such moves are *ask*, where the asking agent expects a response, and *alert*, when the alerting agent expects the other agent to take action since the situation requires action. Consequently, the agent needs strategies to handle the situation when the other agent does not respond as expected.

5.2 Dialogue Actions

The higher-level actions in Figure 2, which relate to different types of dialogues, can be nested in order to meet sub-goals in the process of achieving the overall motive for the dialogue activity defined by the topic. In addition, each nested dialogue is initiated by posing a topic, in the same way as the main dialogue is initiated. As a consequence, the execution of the dialogue body needs to handle the different types of dialogues, motives, their outcomes, the organization of these in the dynamic way, which is needed for the agent to be adaptive and flexible. In this process, some constructs are useful for formalizing the dialogues and their outcomes, and for organizing the process of reaching decisions about e.g, actions to make, and their reasons (arguments). The different types of dialogues, their goals, topics and allowed moves are summarized in Table 2.

The purpose of *argumentation dialogues* is to collaboratively compare different views and generate conclusions about which one is best in a situation. This can be what conclusion to draw, what new knowledge do derive, what action to make, and what changes of mind can be achieved. This is a decision-making approach, which provides strategies for handling conflicting information and views. Information, inference rules and conclusions are typically con-

Table 2 Dialogue types and their characteristics. The topic is drawn from the ACKTUS repositories, using the semantic characteristics of the knowledge nodes.

Type	Goal	Topic	Valid moves
information-seeking	collect information	<i>concept</i>	open, ask, tell, affirm, close
inquiry	create new knowledge in the form of defeasible facts or defeasible rules	<i>i-node</i> (defeasible fact) or <i>s-node</i> (defeasible rule)	open, assert, affirm, close
deliberation	decide about actions to be taken	<i>i-node</i> (defeasible fact) or <i>s-node</i> (defeasible rule)	open, assert, affirm, close
persuasion	change a priority	<i>i-node</i> (defeasible fact) or <i>s-node</i> (defeasible rule), in particular their <i>value</i> as a part of a <i>scale</i>	open, assert, affirm, remind, close
support	enhance human agent's ability	<i>information-node</i>	open, affirm, tell, alert, remind, close

sidered *defeasible*, which means that they can be challenged (*attacked*) and defeated. In our work, we assume that all information is defeasible, and no rules are strict rules, valid in all circumstances. Arguments can be attacked in three ways: on their *premises*, on their *inference* (example in Table 4, row 23) and on their *conclusion* (example in Table 4, row 19). In argumentation literature the notion of *argument scheme* is applied for providing semi-formal or formal templates and defeasible inference rules for different kinds of dialogues [27]. A common example is the scheme "Argument from expert opinion". Sometimes *critical questions* are defined and associated to a particular scheme, which typically are pointers to counterarguments following Prakken's definition [28]. For the purpose of formalizing the different types of dialogues, the concepts *scheme* and *critical question* are applied and modified. It should be noted, that the evaluation of arguments or the application of schemes alone is not sufficient for evaluating a dialogue. The whole process needs to be completed [28]. In the following the common features are described, which will be followed by a number of subsections in which each type of dialogue is defined.

Typically, a formal argumentation framework provides the semantics for solving conflicting arguments for or against a particular topic, regardless what the topic is. The effort to establish an *argument interchange format* (AIF) for exchanging arguments on the world wide web is such example [29]. AIF is presented as an ontology, which takes different types of argumentation schemes, scheme-implementation nodes and their relations into consideration, without relating these to a concept-node system for topics. Most frameworks do also not consider vagueness, or uncertainty, or how these factors may change depending on the context of reasoning. Some theoretical work on argumentation frameworks consider *preferences* and *audiences* (e.g., [30–33]), which relate to the context of reasoning. In the presentation of AIF it is mentioned that the nodes may incorporate values, e.g., representing uncertainty, and domain

knowledge, but not how such information is included in a formal way. In this work, there is a need to take such factors into consideration in dialogues. Therefore, AIF is applied, and extended with domain knowledge in the form of a concept-node system and templates for assessments of features with associated sets of allowed values. The approach to argumentation dialogues is still generic, since the knowledge domain can be replaced, depending on the topic of interest.

The topic of dialogues is retrieved from the ontology incorporating both AIF and the domain knowledge. The domain ontology contains a concept-node system with concepts and their relations. The information-seeking dialogues have as topic a concept, which makes this type of dialogue very generic. Moreover, the domain ontology contains a kind of information node, which has a concept but no values, and functions as the topic for the supportive dialogues (Table 2).

The AIF concepts *i-node* (a different kind of information node, which in this approach is associated to both a concept and a value), and *s-node* (scheme-node, from which rules are generated) are used as topic in inquiry, deliberation and persuasion dialogues, with some differences. The formal distinction made in this approach between the inquiry and deliberation dialogues is that the *claim*, i.e., *i-node*, is in deliberation dialogues associated to a concept related to the node *activity and participation* in the domain ontology, while in the inquiry dialogues the concept is a *thing*. Persuasive dialogues can have both kinds, since the focus is the *evaluation of* the phenomenon represented by the concept. This evaluation is represented by its *value*, which is targeted to be changed.

A *dialogue line* is the sequence of moves conducted by the agents and their time points (e.g., [23]). Such sequence is visible in the example of a dialogue, presented in Table 3 and Table 4. In the following sections each type of dialogue is further described, and exemplified.

5.2.1 Inquiry Dialogues (*wi* and *ai*)

The inquiry dialogue is distinguished from other dialogues in that it is divided into two types following the approach in [23]: *warrant inquiry* (*wi*) dialogue and *argument inquiry* (*ai*) dialogue. The topic of a warrant inquiry dialogue is a defeasible fact, while the topic of an argument inquiry dialogue is a defeasible rule. The purpose of the first kind is to create new knowledge, and in the second to create arguments.

The example dialogue shown in Table 3, starts out as a warrant inquiry (*wi*) dialogue, since the main purpose is to find out whether there is a sleep disorder. The inquiry dialogue evolves when different hypotheses are evaluated and can be assumed to take place after Row 15 in Table 3.

There is a large number of argumentation schemes defined for reaching new knowledge, representing different reasoning strategies and the confidence in the actor, e.g., argument from expert opinion, argument from a position to know, etc [27]. Some reasoning strategy definitions mirror the range of logical

Table 3 Example of different types of dialogues unfolding in a dialogue line. *To be extended with hypothesis evaluation*

t_i	Agent	Statement	Formal representation of a move
1	Rut	Hi Coach, I wonder if not my <i>Sleep pattern may be disturbed</i> .	$(t_1, a_1, open(wi_1, i-node_1))$
2	CA	Hi, Rut. Okay,	$(t_2, a_2, affirm(info-node_1))$
3	CA	Then I would like to talk about <i>sleep</i> .	$(t_3, a_2, open(is_1, concept_1))$ $(t_3, a_2, pause(wi_1, i-node_1))$
3	CA	Did the sleeping pills work for you tonight?	$(t_3, a_2, ask(CQ_1))$
4	Rut	No.	$(t_4, a_1, tell(i-node_2))$
5	CA	Why did you not sleep well last night?	$(t_5, a_2, ask(CQ_2))$
6	Rut	Needed to pee and pain.	$(t_6, a_1, tell(i-node_3)), (t_6, a_1, tell(i-node_4))$
7	CA	Okay, I see.	$(t_7, a_2, affirm(info-node_2))$
8	CA	Now I would like to ask about <i>pain</i> .	$(t_8, a_2, open(is_2, concept_2))$ $(t_8, a_2, pause(is_1, i-node_1))$
9	CA	Where do you have pain?	$(t_9, a_2, ask(CQ_3))$
10	Rut	In my back.	$(t_{10}, a_1, tell(i-node_5))$
11	CA	How severe is your pain?	$(t_{11}, a_2, ask(CQ_4))$
12	Rut	Severe.	$(t_{12}, a_1, tell(i-node_6))$
13	CA	Okay, does not sound good.	$(t_{13}, a_2, affirm(info-node_3))$
14	CA	Let us go back to the topic <i>sleep</i> .	$(t_{14}, a_2, close(is_2, concept_1))$ $(t_{14}, a_2, resume(is_1, concept_1))$
15	CA	How often do you need to get up each night?	$(t_{15}, a_2, ask(CQID_5))$

inference strategies, e.g., deductive and inductive reasoning, causal reasoning etc. Consequently, the agents can apply different strategies, depending on the purpose and quality of the available information. For our purposes, we assume at this point that the agents apply an abductive reasoning method, which includes possibilistic values. The following is the scheme defined for abductive argumentation:

Example 1 Abductive argumentation scheme: F is a finding or given set of facts, E is a satisfactory explanation of F, and no alternative explanation E2 given so far is as satisfactory as E. Then E is plausible, as a hypothesis.

This example applied to our example can be as follows:

Example 2 The human agent is walking around nighttime (F), the human agent's severe pain (E) is a satisfactory explanation of F, and no alternative explanation given so far is as satisfactory as E, therefore, E is plausible as a hypothesis.

In order to execute this type of dialogue, the agent first creates a domain model of how different conditions such as pain or incontinence affect the quality of sleep in general. The following is an example, where $\rightarrow$ is used for representing a defeasible causal relationship: ((pain, severe) $\rightarrow$ (disturbed sleep, probable)). The agent combines this knowledge with the knowledge about the

human agent represented in the user model, which relates to manifested conditions or observations and the user’s priorities: (pain, severe), (walking nighttime), (maintaining pain at endurable levels, highest priority), (maintaining good sleep, highest priority).

However, since there are other potential hypotheses, which may be generated, the Coach Agent needs to continue the reasoning and dialogue in order to reach a satisfactory complete view of the human agent’s situation, following an *exhaustive* method for investigation. The situation is common when there may be more than one hypothesis with same level of satisfaction as the explanation of a finding. Consequently, we will adapt the scheme for allowing the agent work with more than one hypothesis in parallel, and generate an outcome of the inquiry dialogue, which can be utilized as base for further actions. For instance, in the situation when two or more explanatory hypotheses are present with same level of satisfaction, this may lead to the initiation of an information-seeking dialogue, or a deliberation dialogue about what actions to take in order to resolve the situation with conflicting hypotheses, e.g., involving a healthcare professional (e.g., Table 4).

5.2.2 Deliberation Dialogues (dd)

Argumentative reasoning with the goal to decide about what actions to make is often denoted *practical reasoning* in literature (e.g., [21]). The following scheme for practical reasoning is defined by Walton [34]:

Example 3 In the current circumstances R, we should perform action A, which will result in new circumstances S, which will achieve goal G, which will promote value V.

In our scenario, this may correspond to the following:

Example 4 We know that the pain is severe (R), and if the pain is reduced (A), this will result in the new circumstance where the pain is mild (S), which will achieve the goal to keep the level of pain at a manageable level (G), which promotes good sleep (V).

The execution of this type of dialogue follows the same procedure as in previous example. The topic of the dialogue is *reduce pain*, which is an action (Table 1). The agent creates a domain model of how pain and other conditions and medication may affect the quality of sleep in general: ((pain, severe) → (disturbed sleep, probable)), (painkiller → reduced pain). The agent combines this knowledge with the knowledge about the human agent represented in the user model: (pain, severe), (walking nighttime), (maintaining pain at endurable levels, highest priority), (maintaining good sleep, highest priority).

The dialogue in this situation will deal with what to do to reduce the pain level, in order to affect the sleep disturbance in a positive way. In our example, using painkiller to reduce pain would be the suggestion (rows 21-25 in Table 4).

Table 4 Brief example of persuasive and deliberation dialogues. *To be extended with support dialogues*

14	CA	Let us go back to the topic <i>sleep</i> and <i>priority</i> .	$(t_{14}, a_2, close(is_2, concept_1))$ $(t_{14}, a_2, open(pd_1, i-node_7))$
15	CA	You said that maintaining good sleep is of highest priority.	$(t_{15}, a_2, remind(i-node_8, \{\}), a_1)$
16	Rut	Yes.	$(t_{16}, a_1, affirm(info-node_4))$
17	CA	You also told earlier that maintaining pain at endurable levels is not important.	$(t_{17}, a_2, remind(i-node_7, \{\}), a_1)$
18	Rut	Yes.	$(t_{18}, a_1, affirm(info-node_4))$
19	CA	Since it is likely that severe pain causes disturbed sleep, and you have severe pain, and you think maintaining good sleep is of highest priority, then it should be important to maintain pain condition at an acceptable level. What do you think?	$(t_{19}, a_2, assert(i-node_9, \{s-node_1, i-node_6, i-node_8\}))$ $(t_{19}, a_2, ask(CQ_5))$
20	Rut	Okej, important I guess.	$(t_{20}, a_1, affirm(info-node_1))$ $(t_{20}, a_1, assert(i-node_9, \{s-node_1, i-node_6, i-node_8\}))$
21	CA	Okej, let us talk about what to do about the <i>pain</i> .	$(t_{21}, a_2, affirm(info-node_1))$ $(t_{21}, a_2, close(pd_1, i-node_7))$ $(t_{21}, a_2, open(dd_1, i-node_{10}))$
22	CA	Since taking painkiller typically reduces pain, then you can take painkiller. What do you think?	$(t_{22}, a_2, assert(i-node_{10}, \{s-node_2\}))$ $(t_{22}, a_2, ask(CQ_5))$
23	Rut	No, painkiller does not work.	$(t_{23}, a_1, tell(i-node_5))$ $(t_{23}, a_1, assert(i-node_{11}, \{s-node_3\}))$
24	CA	Okej, I see. Do you want to talk to your nurse about medication?	$(t_{24}, a_2, affirm(info-node_2))$ $(t_{24}, a_2, ask(CQ_6))$
25	Rut	Yes.	$(t_{25}, a_1, tell(i-node_{11}))$
26	CA	Okej, don't forget to talk to the nurse about your medication!	$(t_{26}, a_2, affirm(info-node_1))$ $(t_{26}, a_2, remind(i-node_{11}, \{s-node_3, s-node_1, i-node_6, i-node_8\}), a_1)$ $(t_{21}, a_2, close(dd_1, i-node_{10}))$

5.2.3 Persuasive Dialogues (*pd*)

Persuasive dialogues aims at resolving a conflict of opinion [28].

To formalize this type of dialogue, the agent creates a user model of the human's prioritized activities, for example: (taking medication, important), (wellbeing, very important) and (maintaining good sleep, highest priority). For illustrating a persuasive dialogue, the user's preference regarding managing pain is set to the the following: (maintaining pain at endurable levels, not important). Then the agent combines this knowledge with the knowledge obtained from the human actor during the dialogue. Suppose the human has

disturbed sleep and initiates a dialogue with the Coach Agent about the topic "sleep patterns may be disturbed". During the information seeking dialogue, the agent asks the human about pain and if the human responds that a pain condition is present, and the pain is "severe", then the agent updates its user model with (pain condition, yes) and (pain, severe), and initiates a persuasive dialogue (rows 14-20 in Table 4) based on its knowledge about the relationship between sleep and pain condition.

The agent makes new statements, reminds the human about the relationship between pain and sleep, and the human changes the evaluation of the importance of managing the pain condition (the value *important* is stronger than the value *not important*). Next step for the agent is to propose actions to do something about the situation.

5.2.4 Information-Seeking Dialogues (*is*)

The topic of an information-seeking dialogue is a *concept*, which is a broader topic than the defeasible facts or rules, used as topics for an inquiry, deliberation or persuasion dialogue. An information-seeking dialogue typically unfolds as an interview, where the Coach Agent in our examples asks relevant questions to the human agent, and receives answers. To some extent the agent evaluates the new information, however, primarily for deciding upon next step, typically what question to ask next. Therefore, we use the move *tell* in the information seeking dialogues instead of *assert*, in order to distinguish between answering questions for reasoning purposes such as in inquiry dialogues, and answering primarily for collecting information.

5.2.5 Support Dialogues (*sd*)

The topic of a supportive dialogue is an *information-node*, which is not associated to a value. The content of a support dialogue is typically the outcome of an earlier conducted deliberation dialogue, where the human agent and the Coach Agent have agreed upon a plan of actions to be conducted. The actions performed as support dialogues are one of the following: provide the human agent with information or advice, remind the person of actions to make, and alert the person when important things need to be done. A remind and alert move are arguments, which contain the information about what is to be done, i.e., a claim, together with the motivations, i.e., the grounds, which support the claim.

6 A Model of the Coach Agent

For the Coach Agent to be able to conduct dialogues similar to natural dialogues, it was necessary to design the agent as an autonomous actor, with its own knowledge base, goals and priorities. There is for instance, behavioral knowledge, which is generic regardless which type of dialogue or topic is at

focus. Since this knowledge needs its own semantic model, yet related to the human actor's, the Coach Agent was modeled as a separate project in ACKTUS (Figure 1). As a consequence, the Coach Agent shares the core ontology with the human agent and the domain models. This function as the common vocabulary in dialogues.

The Agent model contains structures, which function as tools for the Coach Agent to use in the adaptation of its behavior. These tools aids the agent in the *conceptualization, knowledge acquisition and reasoning*, related to cooperative human-agent activities.

The Coach Agent associates each action with a concept that is common for both the human and the agents such as *starting, sustaining* and *ending a conversation*. These concepts are used to define certain behaviors that are typically appropriate in a phase of a dialogue. For example, *"Hi"* is associated to concept *"starting a conversation"*. The different types of dialogues and dialogue moves described in previous section are also integrated in the agent's model of dialogue activity. For instance, the move *affirm*, means that the agent emphasize and affirms the emotional state, which the human agent expresses, by using different empathic statements such as *"does not sound good"* and *"I see"*. This enables the Coach Agent to simulate the behavior of a participant who listens with empathy. The agent shares also the same conceptual model of entities as the human actors, which are relevant to a human actor based on the International Classification of Function, Ability and Health (ICF)¹.

The Coach Agent acquires new knowledge partly by by posing explicit questions. One example is in the situation when the Coach Agent presents a claim about a topic, followed by the question *"What do you think?"* (e.g., Line 19 and 22 in the dialogue example in Table 4). This is a generic question, which gives the agent some indication if the human agent agrees to the claim or if the agent needs to adjust its strategies for the next move in the dialogue. The question about perceived appropriateness of a question, which the agent asks, and which was used in the evaluation study is also retrieved from the agent model (Section 7).

The agent also retrieves formal models from the agent domain repository, which can be used for reasoning, in the same way the agent retrieves models from another domain repository in the dialogues about a topic.

7 Pilot Evaluation Study Results

The main research questions addressed in the evaluation study is related to the overall idea of a dialogue system for supporting everyday issues, how a sense-making dialogue would unfold and what topics would be interesting to elaborate upon. Other research questions are related to interaction design of the dialogue system. Since the evaluation study was limited to a subset of the models described in this article, and the study sample was small, the results are only indicative, aimed to inform further development.

¹ <http://www.who.int/classifications/icf/en/>

A purpose was also to evaluate the method to allow the user to evaluate the appropriateness of each of the Coach Agent’s statements within its context where it was stated. A question about the appropriateness of the agent’s behavior was associated to a four-item Likert scale with the values *inappropriate* (1), *somewhat inappropriate* (2), *somewhat appropriate* (3) and *appropriate* (4). The 13 participants’ interaction with the dialogue application was logged, and resulted in a corpus with a total of 28 dialogues, 156 complete turns, and a total elapsed time of 2 hours (approximately 10 minutes per participant). The examples from the dialogues presented in this section have been translated from the participants’ native language into English.

The results are divided into results related to the purposes of dialogues, the appropriateness of the agent’s behavior and interaction design in the following sections.

7.1 Purposes of Dialogues

The dialogues were based on the fictive persona Rut’s health and priorities. Most of the participants could personally relate to the dialogue topics, a few referred to a family member. In general, the available dialogue topics were considered interesting and relevant.

A generic attitude among the participants were that there is need for supportive dialogue systems, on various themes. Additional health-related topics, which were suggested were topics related to eating habits and eating disorders. Besides having healthcare issues to discuss, some pointed out the need in particular situations, when getting lost in the forest, or misplaced a car in the forest or bike in the city center. Others pointed out the potential benefits relating to getting access to social networks, having dialogues about societal issues such as politics, weather, golf and other sports.

A few expressed enthusiasm and curiosity about the idea and wanted to use the dialogue system merely for the fun of it: "...this is fun, let me try another one!"

Two of the older adults expressed skepticism, and did not see the point in using the dialogue system. One of these had also some difficulties using a smart phone, which may indicate a threshold for using new technology in general. By contrast, the other skeptical older adult was already using alternative ways to accomplish potential dialogue purposes, by using search engines for health issues, GPS for navigating, etc. Consequently, the group of participating older adults, including the two skeptical persons, illustrates the broad heterogeneity of the group of older adults.

The therapists highlighted the potential benefits of using a proactive human-agent dialogue system for providing active support for improving strength and balance, e.g., by supplementing the dialogues with sensor information from daily activities and physical exercises performed, analyzed over time.

7.2 Perceived Appropriateness of the Agent's Moves

One of the therapists who tested the system perceived the approach to evaluate the moves within the context it occurs, as highly beneficial. She compared to structured clinical interviews, where it is typically not known how the patient perceives the appropriateness of the question asked.

The question, which was used for evaluating the appropriateness of the agent's moves, was found to be used in two different ways. Based on this observation, we distinguish between two types:

1. *Context-related appropriateness*: the appropriateness based on the immediate context of the agent's move, e.g., placement in the dialogue line, and
2. *Topic-related appropriateness*: appropriateness related to the topic of the dialogue.

The moves, which were categorized as the first kind and valued inappropriate to some level, were considered a failure of the agent, and are aimed to be minimized by improved strategies for the agent to use when choosing behavior. Consequently, these were removed, when analyzing the second kind, which was considered relating to the domain knowledge and not behavior issues.

7.2.1 Context-related appropriateness

Regarding the context-related appropriateness, some of the agent's moves were considered as inappropriate to some extent due to the agent's inadequate understanding of the human's responses. The agent needs to understand the context of human's response, update its belief base and act accordingly. Therefore the agent needs strategies for wisely selecting moves based on the human's response and the unfolding of the dialogue line. For evaluating the agent's improvement of behavior for future studies, a Contextual Appropriateness Value was defined and tested on this data sample. The total contextual appropriateness value for the complete corpus was 90.4%, and consequently, the error rate was 9.6% (16 of 167 moves) (Table 5).

7.2.2 Topic-related appropriateness

To what extent the agent's moves were perceived by the participants to be relevant to the overall dialogue topic was analyzed. The agent uses the domain knowledge, which it retrieves from a domain knowledge repository modeled by domain experts, to build a model of the knowledge domain, related to the selected topic of a dialogue. The domain experts have in their modeling created a knowledge model with interrelated topics following their view on to what extent different phenomenon are relevant to each other. However, human individuals who have a dialogue with the agent may perceive the level of appropriateness differently. We evaluated this topic-related appropriateness based on the participants' answers to the question about appropriateness, and mapped them to the concept associated to the content of the agent's move.

Table 5 Total number of dialogue turns (Turns), number of Contextual Appropriateness Errors (CAE: turns, which are to some extent inappropriate), and mean of errors

User	Turns	CAE
1	13	1
2	13	1
3	8	1
4	9	4
5	13	1
6	28	0
7	28	3
8	9	2
9	9	0
10	9	1
11	8	1
12	9	1
13	11	0
Total	167	16
Mean	12.8	1.2

The most common topic, which the participants selected was the topic *sleep patterns may be disturbed*. The corresponding concept is *sleep*. Table 6 provides a summary of the participants’ responses about appropriateness of different moves in relation to sleep. After removing the contextual errors and missing information (marked as not available (*n/a*) in Table 6) , we can observe that most of the turns made by the agent were considered as appropriate to the topic sleep.

7.2.3 Discussion

To summarize, identifying the topic-related appropriateness of participant’s moves is to large extent dependent on the perspective of an individual. Some participants considered the questions related to pain and the presence of worries as appropriate while some others considered the same topics as inappropriate to different extent. Some comments by the participants indicated that they associated these questions to their own situation and health conditions, rather than the persona Rut’s situation, which was the reason for why these moves were perceived as inadequate. Consequently, if the participants had been able to create their own profile in the system for the agent to build a user model from in the dialogues, then some of these less appropriateness evaluations had been different.

The question about reasons for why getting up at night had a broad range of answering alternatives, among which a few participants found less relevant alternatives.

The overall goal of our human-agent dialogue system is to minimize the contextual appropriateness errors and maximize the topic-related appropriateness. To attain this goal, future work includes developing strategies for the agent to wisely select moves based on changes in its belief base. Moreover, these improvements will be evaluated with a group of older adults who create

their own profiles in the system, for evaluating to what extent the perceived inappropriateness is related to the domain knowledge, and the differences in the view on this. If the differences are significant, a machine learning method will be built for the agent to learn how an individual value the topic-related appropriateness and adapt its selection of moves to the individual.

Table 6 Perceived topic-related appropriateness: inappropriate (✗), somewhat inappropriate (✗), somewhat appropriate (✓) and appropriate (✓). The abbreviation (n/a) (not available) represents missing information where a particular question was not asked to the user or the user chose not to give feedback.

Ques \ Users	1	2	3	4	5	6	7	8	9	10	11	12	13
Sleeping well all night	✓	✓	✓	n/a	✓	✓	✓	✓	✓	✗	✓	n/a	✓
Cause of getting up at night	✓	✓	✓	✓	✓	✗	✓	✓	✗	✓	n/a	✓	✓
Severity of pain	✓	✓	✓	n/a	✓	✓	✓	✓	✓	n/a	n/a	✓	✓
Having worries	✗	n/a	n/a	✓	n/a	n/a	✓	✓	n/a	n/a	n/a	✓	n/a
Severity of worries	n/a	✗	n/a	✓	✓	n/a	✓	n/a	n/a	✗	✗	n/a	✗
Taking some medication	✓	n/a	✓	n/a	✓	✓	n/a	n/a	✓	✓	n/a	n/a	✓
Handling medications	n/a	n/a	✗	✓	n/a	✗	✓	✓	✗	n/a	n/a	✓	✗

To summarize the evaluation of the described method of valuing the appropriateness of the agent’s behavior within its context where it occurs, the approach was found to add significant value to the evaluation study, without disturbing the participant from participating in the dialogue. However, considering the limitations of the pilot study, and the two possible ways to interpret the question, the method will also be further evaluated in future studies.

7.3 Interaction Design

The older adults did not comment on the visualization of dialogues in the graphical interface. By contrast, the therapists put significantly more interest in how the dialogues would appear to the user. They commented on what parts of the dialogue line were visible at different phases in the dialogue, the unfolding of dialogues, on how the responses would be given by the user, etc. Their main concern was how reducing the amount of information presented to the user, while not losing the information about in which context the dialogue was situated. The interface design will be improved based on their comments and suggestions.

One therapist mentioned the concept of an "avatar", sometimes used for representing an Embodied Cognitive Agent (ECA). She viewed the Coach Agent as such, and discussed how body language could be added to the static figure used in the graphical user interface for representing the Coach Agent.

Other comments related to the benefits of, and how mediate, the dialogues through spoken language instead of text. Testing voice-based dialogues will be a natural step towards a more adaptive human-agent dialogue system, considering the large proportion of older adults who have sight impairments. One obvious benefit from using the ontology-based semantics of dialogues, was illustrated in the pilot evaluation study, since it allowed to apply the two languages needed by the participants. This will also facilitate the inclusion of additional modalities in the interaction design, which will be evaluated in future studies.

8 Conclusions

The purpose of this work was to define a generic model of purposeful human-agent dialogues about health-related topics. This was done based on analyses of scenarios, personas and models of human behavior. The resulting conceptual model includes four models to be included in a software agent's belief base; i) a user model, ii) a model of the domain knowledge related to the topic of the dialogue, iii) an agent model, and iv) a dialogue activity model. The major contributions of this work are the agent model and the dialogue activity model, the latter to be shared between the human and software agents. The models were implemented into a prototype system for human-agent dialogues, which was evaluated by therapists and a group of older adults.

Future work includes the development of the dialogue and reasoning strategies for the agent to improve its ability to adapt to the individual and the situatedness of contextual and natural dialogues.

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